

AR Magic Lantern: Group-based Co-Located Augmentation Based on the World-as-Support AR Paradigm

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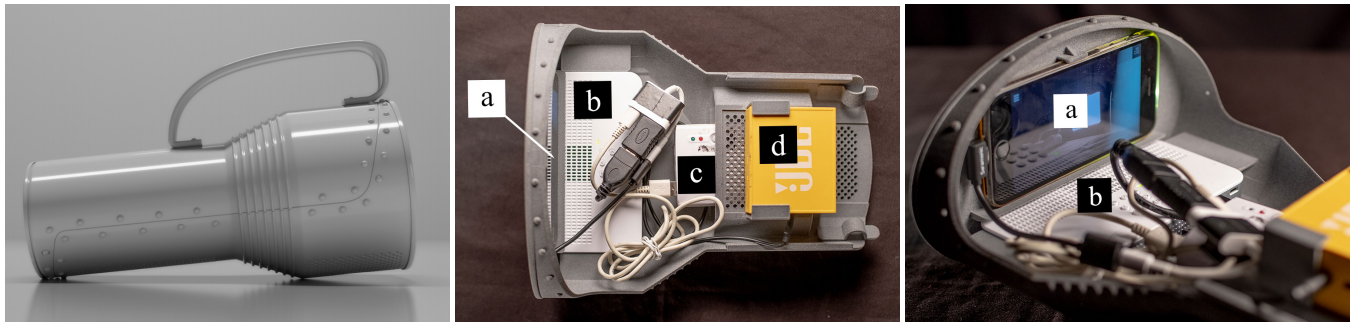


Figure 1: (i) The external casing of the AR Magic Lantern. (ii & iii) Components of the magic lantern: (a) Apple iPhone 7 smartphone, (b) a Laser Beam Pro C200 pico-projector, (c) a USB battery pack, and (d) a JBL Go portable speaker

ABSTRACT

We present the AR Magic Lantern, an augmented reality flashlight that enables its holders to see and interact with situated virtual content that it projects onto the physical surfaces around them. The AR Magic Lantern tracks its position and orientation, so the projected content is truly a part of the physical space and is thus relevant for all members of a group. This makes it especially suitable for implementations of the World-as-Support AR paradigm, where augmented experiences should be shared and situated. We contribute the design principles and methodology used in the development of the hardware and software of the AR Magic Lantern. We also describe the interactive AR experience that we developed for a local cultural heritage site and which will be demoed during CHI 2022.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**; *Mobile devices*; *Interaction design theory, concepts and paradigms*; *Displays and imagers*.

KEYWORDS

augmented reality, mixed reality, spatial AR, AR interaction paradigms, full-body interaction

ACM Reference Format:

Paul Hine, Loza Tadesse, and Narcis Pares. 2022. AR Magic Lantern: Group-based Co-Located Augmentation Based on the World-as-Support AR Paradigm. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3491101.3519918>

1 ADDRESSED PROBLEM AND SYSTEM

Augmented Reality (AR) has the evocative power of allowing us to see that which is invisible to the naked eye and add new layers of meaning to our (physical) world. Although the community of people related to HCI and digital media have a relatively good idea of what AR is and how it can be used, in our experience, society at large does not have such a good notion of this medium. Therefore, if we want to introduce it to public spaces and foster its introduction to a number of application areas such as Education, Cultural Heritage, Tourism, etc, we must not only make it available to them (easily findable and affordable), but we must also provide them with an interaction paradigm that is natural and easy to adopt. Moreover, once we have a good paradigm, we must provide an adequate device that allows that paradigm to be implemented in applicable contexts.

1.1 Related Technologies and Our Device

Current AR technologies are either based on headsets or on smartphones. Headset-based AR, via see-through head-mounted displays, was an adaptation of existing military aircraft Heads-Up Displays (HUDs) [4]. They implicitly defined a paradigm by which users acquire a special capability of their eyesight by wearing a special set of glasses, such as Microsoft's HoloLens [10]. On the other hand, a much cheaper and readily available AR technology has been successfully used in many application areas for the general public:

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CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3519918>

smartphone-based AR. Thanks to the high degree of integration, exponential increase in computing power, and capabilities provided by special hardware sensors (high-res cameras, digital compasses, accelerometers), smartphones and tablets can effectively overlay digital stimuli over a captured image of the physical world and show it on their screens. This technology came with a paradigm of its own, known as “Window-on-the-World” (WoW) [11]. Both technologies have a number of drawbacks when used in public spaces, addressed to the general (inexperienced) public, and applied to group experiences (which are the most usual types of experiences in public spaces).

In the past years, a number of other technologies have evolved and merged to allow for a different type of augmentation. On the one hand, “projection mapping” has evolved to produce very impressive, entertaining, and educational experiences by projecting digitally controlled stimuli over physical space [3, 5, 8]. On the other hand, projection devices have evolved into very small form factors and low power consumption. These “pico-projectors” provide the possibility of easily moving the projection to any space and have opened the door to new directions in interface and interaction design [7]. Both have merged into what is known as projective augmentation or *Spatially Augmented Reality* (SAR) [12].

We present a new device, the AR Magic Lantern (ARML), based on a relatively new AR paradigm, the World-as-Support (WaS) [9]. This device intends to improve interactive augmented experiences for small to medium-sized groups (two to eight participants) using an evolved, situated version of SAR and a very natural and useful interaction metaphor: the flashlight (torch). Hence, our design process aimed to produce a device whose physical appearance was clearly that of a flashlight so that it instantly evoked the metaphor of its handling and use (figure 1). Devices that help us light our way through dark places are ever-present in popular culture and knowledge of their affordances continues to be passed on to subsequent generations. When transposing these affordances into the realm of HCI, they provide an adequate “cultural form” as posed by Horn [6]. This combination of paradigm, metaphor and device provides clear improvements over the aforementioned drawbacks presented by the existing AR technologies as we will detail in later sections.

Our ARML prototype consists of an (a) Apple iPhone 7 smartphone, (b) a Laser Beam Pro C200 pico-projector, (c) a USB battery pack, and (d) a JBL Go portable speaker (figure 1). The smartphone provides HDMI output and accepts power from an external battery pack. The HDMI output from the smartphone sends audio and video to the mini-HDMI input of the pico-projector. The audio output of the pico-projector is connected to the audio input of the portable speaker. The smartphone is running our custom application built in Unity that takes advantage of the Apple ARKit SDK to perform the necessary SLAM (simultaneous localization and mapping) and association of digital 3D content with virtual anchors attached to physical locations. The application also manages the rendering distribution between the primary display of the smartphone and the pico-projector, which serves as a secondary display.

1.2 Solving for Co-Located Group-oriented AR

Despite the interesting technological potential provided by existing AR technologies (headset-based and smartphone-based), they

present some serious drawbacks when used in public spaces and applied to groups of visitors. Malinverni et al. [9] saw the need to formalize a new paradigm to conceptually guide technological development based on projective augmentation, the “World-as-Support” (WaS), which proposed two main conceptual approaches: (i) Use the physical world as a physical support to project the digital stimuli that augments it, and (ii) use the meaning of the physical world (of the physical geometry, architecture, and objects, but also of the cultural and social context, the historical storyline, etc.) as the baseline meaning that will support the meaning of the augmented stimuli.

According to Malinverni the WaS paradigm seems to provide a number of interesting advantages over the “Window-on-the-World” (WoW) paradigm defined by smartphone-based AR when deployed in activities in which groups of users must keep a sense of group and not feel isolated. Some of these properties are also advantages over the headset AR configurations. The following are the advantages of WaS in the context of groups of users in public spaces and how we have dealt with them in our ARMLs:

Group User Experience: The User Experience (UX) with ARMLs is immediately open to the group of people that accompany the *device-holder*. The projection of the visual stimuli onto the physical surroundings allows every group member to immediately experience and comment on what is being shown. This creates a bond between the members of the group, allows rich discussions, and provides opportunities for different points of view and personal backgrounds to emerge in the conversation. This is not easy (or often even possible) in smartphone or headset-based AR, since the focus on the small screen of the device and the closed personal display, respectively, isolate the users from the rest of the group. Moreover, both technologies force all members in the group to use one such device if they wish to keep within the global augmented experience. However, some members of a visiting group may not want or cannot undertake using such devices; e.g. senior members may not feel comfortable using technologies, or people with special needs or disabilities may need help. ARMLs do not force all users to hold a device.

User Interaction: The interactive aspects of the UX with our ARMLs are also open to the *non-holders*. When a device holder points the ARML at a virtual object that is “grabbable,” this is detected through a ray-casting method. Once the intersection of our ray cast and the object is verified, a visual timer appears and slowly advances, providing a sense that once the time elapses then something will happen (also known as a “Dwell click”). The device-holder may cancel “grabbing” action by pointing away from the object. If the time does elapse, the virtual object is “grabbed” and carried within the field of view of the projection wherever the user now points. To “release” the grabbed object, the user must find a location where it is relevant to do so. Our ray casting algorithm detects the correct location and a new timer is activated to give time to the users to decide whether they wish to release the object or not. Grabbing and releasing objects can lead to: relocation of objects; assembly of objects from parts; construction of structures; combination of objects to generate new objects (as in cooking); etc. This interaction is kept open to the group as everyone may see what is going on. This is not the case of the smartphone (tapping or dragging on the small screen), or the headset (gestures detected by the headset).

Moreover, ARMLs open the potential to let the non-holders interact with the augmented stimuli. When a non-holder stands between the ARML and a projected object, the device can detect the human figure and track its skeleton. Through this tracking, the device can allow the non-holder to use natural gestures to manipulate some of the virtual objects, providing a huge potential to design embodied interaction for both device-holder and non-holder. This is not possible with smartphones or headsets since the non-holders cannot see the effect of their actions.

Situated Action: smartphones place the main focus of interest (of the device-holder) on the visualization screen; what a number of researchers have come to call a “digital divider” [2]. That is, although an AR experience is meant to augment physical space, smartphone AR undermines augmentation because the whole experience is screen-based, providing a weak situatedness. In fact, the importance of being on-site while having the AR experience is diminished as smartphone users could have stayed comfortably at home and used their devices to visualize 360° images of the physical site with overlaid virtual objects as they spin around. This is also the case of headsets that are not see-through where the experience could well be a full VR experience. However, ARMLs place the focus of interest of the users (both device-holders and non-holders) on the physical space first and foremost. ARMLs make no sense unless users are located on site at the specific location.

Meaning: From a communicational standpoint, the medium defined by ARMLs is taking most advantage of the meaning that is found at the physical location. By “meaning” we may refer to the definition of a context, a story, a cultural rite, a social convention, etc. Smartphone AR digitizes the physical world through a camera and mixes it with the virtual content to be displayed on the screen loosing its hierarchical structure where the physical space should be the main source.

Safety: ARMLs impose no viewing range constraints and users can be attentive to any dangers related to terrain relief or objects that can get in the way. Smartphone AR captures user focus on the screen and places them at greater risk of hurting themselves or damaging protected artifacts and structures (as evidenced by the popular smartphone-based AR app *Pokemon Go* [1]). The headset is: invasive, which many users do not like; cumbersome, despite recently becoming lighter and wireless; and has hygiene issues, which can be especially difficult to deal with in public spaces, and have become a real headache with the COVID-19 pandemic. In contrast, our handheld ARMLs are non-invasive and may be easily sanitized when handed from one user to the next.

1.3 Description of Exhibited Interactive Experiences

Heritage sites are excellent test grounds for AR because they provide a rich historical context but are almost always devoid of their original content; i.e. objects in general (furniture, pottery, tools, art, etc) are usually taken to museums for storage and protected display. This often makes the experience of the visitors of heritage sites very abstract. Our system restores the objects to their original locations, makes the visitors aware of the human (social and cultural) activity that took place in those locations, places them at the center of the

action, and challenges them to perform activities to learn about them “in the flesh.”

Thanks to a research collaboration with MUHBA (the Museum of the History of Barcelona), we have been able to design and test three different interactive experiences for our AR Magic Lanterns at the heritage site of *Barcino*, the ancient Roman Barcelona from 2000 years ago. We have adapted one of these experiences, *The Garum Factory* (described below), into a portable demo that can be presented in spaces outside of the museum context.

1.3.1 Interactive Experience: *The Garum Factory*. A large section of the *Barcino* excavation is occupied by the remains of a facility where the Romans produced *garum*, a popular and expensive sauce or paste made primarily from fermented fish, and which they spread on bread or other food. The remains include the room where they used to chop the fish to prepare them for fermentation, as well as a few rooms full of *dolia*, large pottery spherically shaped containers, where the sauce was mixed, prepared, and stored. We developed an activity that helps visitors to the ancient *garum* factory understand the process and facilities involved by letting them collect and combine the ingredients to make the fishy sauce. First, the device-holder is directed (e.g. by a museum guide) to point the ARML at a stone column that once formed part of the factory. There, they find simple instructions that list the “Ingredients Missing” to make the sauce: salt, aromatic herbs, and fish (figure 2a). As the device-holder explores the space of the factory with the ARML, they will find these ingredients, e.g. the fish on the floor of the chopping room, the aromatic herbs and salt in different *dolia*. If the device-holder “grabs” one of these ingredients by aiming the ARML at it until the visual timer elapses (figure 2b), they now hold the ingredient and must find a place to drop it. A visual sparkle effect in another *dolium* helps alert the device-holder that it might be a place to do so; also, when they pass the ARML over that spot, the timer appears to let them know it is a suitable place for dropping the ingredient. When they drop the ingredient in that *dolium*, a sound plays from the lantern and the “instructions wall” updates by removing that ingredient from the list of “Ingredients Missing.” When all three ingredients are dropped in the mixing *dolium*, they will see an animation of the ingredients mixing together and the sauce bubbling as it ferments. Accordingly, the “instructions wall” updates to “Congratulations! Fermentation time: 2 weeks.”

2 RELEVANCE TO CHI COMMUNITY

2.1 Novel Device and Open Source Technology

We provide the design steps taken to define and prototype a device that we have called the AR Magic Lantern, and we provide a running demo of a subset of experiences that we have designed for the *Barcino* heritage site. This device has been tested with real users and shown it works and that it does in fact implement the WaS paradigm.

We also plan to open source the foundation libraries of the ARML device in the near future, inviting a large community of researchers and developers to help advance the functionality and robustness of the system. We want to migrate from the present dependency on the iPhone to custom hardware that is running an open OS such as Linux. We also wish to define a powerful editor and content

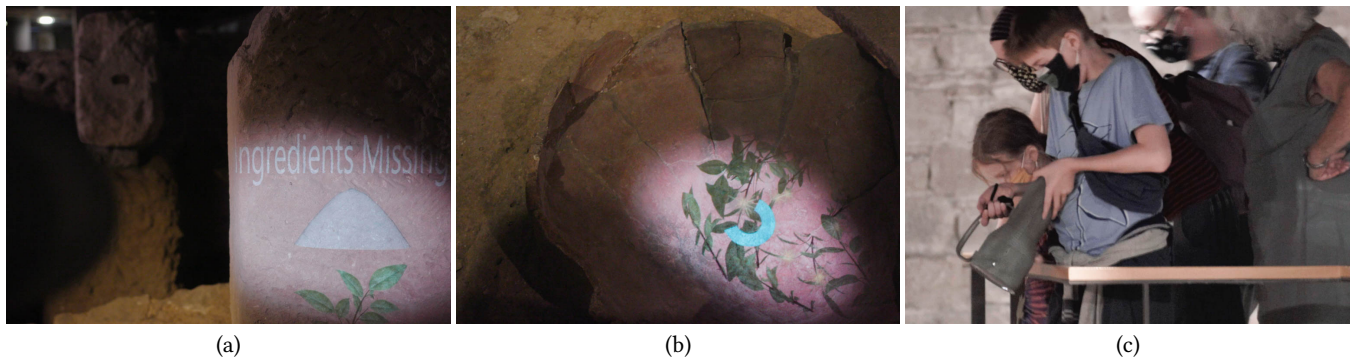


Figure 2: The Garum Factory augmented reality activity. Pointing the ARML at a stone pillar in (a) shows the "Instructions Wall" for the activity. Pointing the ARML at the leafy herbs in (b) starts a visual timer in the beam that tells the user that an interaction is possible with this item. In (c), real museum visitors use the AR Magic Lantern during the trials for the preliminary studies

manager so that non-programmers like museum curators, archaeologists, managers of heritage sites, historians, etc., can construct their own experiences in an easy and powerful way. Finally, we want to further extend the hardware integration and work with fabrication partners to produce and distribute the ARML at a cost that is viable for museums, other cultural heritage sites, schools, etc.

2.2 Interaction Design for WaS AR

We tested these interactive experiences with a number of randomly-selected visitors to the Barcino heritage site. Most of these visitors were in a party of two, although we had some in larger parties of three, four, or even five. Those in parties of two were asked to act as device-holders for part of the experiences and as non-holders for the other part. In larger parties we asked only two of them to act alternately as device-holders and non-holders, while the rest acted as non-holders for the total duration of the trial.

We also developed the same AR experiences in the WoW paradigm to be able to compare user experience and reactions. This means that rather than using the ARML, the visitors used the typical smartphone-based AR that is relatively well known nowadays. We undertook the WaS and WoW trials on different days to prevent users of one paradigm from glimpsing anything whatsoever of the other paradigm. In this way, we also avoided a bias between the two experimental groups.

There were some interesting reactions that we now describe as anecdotal but nonetheless provide useful insight for AR design. One of the first plainly transversal reactions was how non-holders in the WaS paradigm (using the ARML) start to immediately comment on what is being projected by the device-holders, showing clearly that the experience becomes shared and embraced as a group (figure 2c). This seems to suggest that the adequacy of the WaS paradigm for groups is indeed a value of this paradigm and device option, since this was not observed for the WoW paradigm (using smartphone AR). This was even more obvious with larger groups of users in which all the members in the party would join in the discussion of the experience they were sharing through the projected images thanks to the ARML.

There was also a notable difference in the physical behaviour between participants in the smartphone WoW trials and those in the ARML trials, especially in the non-holders. The participants not holding the ARML were more prone to stay involved in the action of the activity by physically following the device-holder around the space and keeping their eyes on the projections. Even when there was nothing being projected, they stayed involved by looking around the area and gesturing vocally and physically to suggest that the device-holder try pointing the ARML in a new direction. This attention from the non-holder was true not just for pairs of participants but also in larger groups, where every member of a family of four or more, for example, would follow the lantern beam with their eyes and their feet (figure 2c).

By contrast, we observed that the physical involvement by the non-holder was more limited in the WoW trials. In larger groups the non-holders would wander off to a different part of the exhibit space, e.g. the parents letting their children partake in the activity on their own. When the non-holders did remain near the visitor holding the smartphone, their eyes were always on the screen. In fact, while the non-holders did sometimes try to suggest an action to the device-holder, they used far less body language to do so, and the gestures they did use were always aimed at the screen and never at the physical space around them. This supports the hypothesis that the entirety of the WoW experience is limited to the frame of the screen. The form factor and affordance of the screen also seemed to invite participants to try tapping or dragging the virtual objects they saw on it, even though it had been explained and demonstrated that the actions were triggered using the visual timer (identical to the ARML). Finally, all participants in the WoW trials, holders and non-holders alike, seemed to physically move about the exhibit space much less than those in the WaS trials. Instead, they tended to stay planted in one spot and only move the phone around.

Hence, the study clearly supports our hypothesis that WaS promotes rich, embodied, spatial exploration, as well as group cohesion, collaboration, and shared experiences for holder and non-holder users alike.

ACKNOWLEDGMENTS

Part of this research has been funded by the Culture Institute of the City Hall of Barcelona (ICUB, Institut de Cultura de Barcelona). The industrial design of the casing of the ARMLs was undertaken by Victor Freire. Hewlett Packard Spain (in Barcelona) kindly sponsored the 3D printing of the new casings in their Multi Jet Fusion powder 3D printing system. We specially thank the fruitful collaboration with and help from the Museum of the History of Barcelona (MUHBA, Museu d'Història de Barcelona).

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